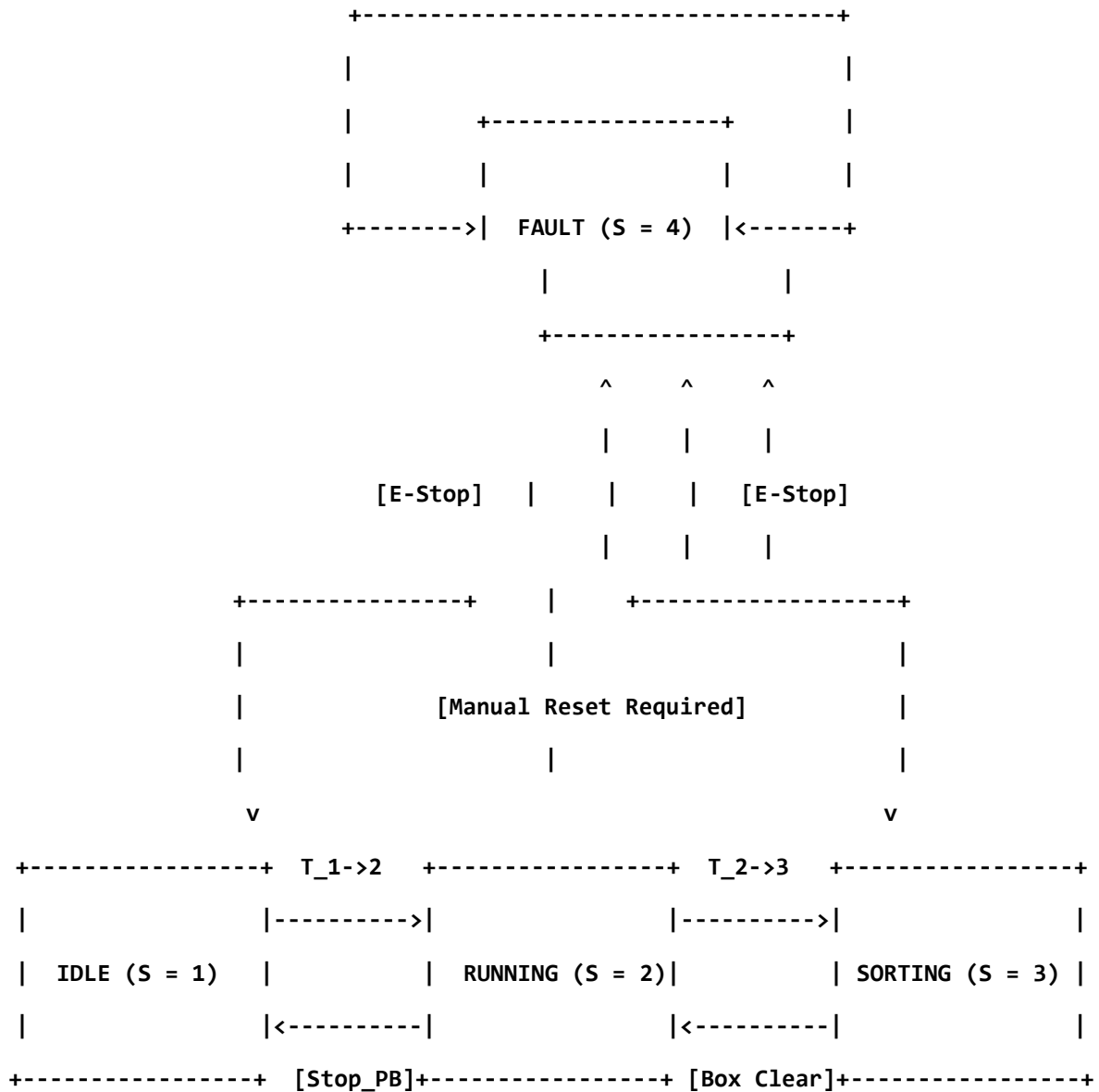


# PROJECT 4: PLC-BASED CONVEYOR SORTING SYSTEM



| Transition Tag         | From State           | To State             | Engineering Logic Condition                                                                               | Practical Action on the Factory Floor                                                            |
|------------------------|----------------------|----------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $T_{1 \rightarrow 2}$  | <b>Idle (S=1)</b>    | <b>Running (S=2)</b> | i_Start_PB AND NOT(i_Stop_PB) AND i_EStop_Mon AND NOT(b_EStop_Fault)                                      | Main conveyor belt drive contactor is instantly energized and downstream sensing loops activate. |
| $T_{2 \rightarrow 3}$  | <b>Running (S=2)</b> | <b>Sorting (S=3)</b> | i_Prox_Sensor AND NOT(b_EStop_Fault) AND NOT(b_Batch_Complete)                                            | The belt maintains speed while the sequence-based On-Delay Transit Timer (TON) initiates.        |
| <b>Automatic Reset</b> | <b>Sorting (S=3)</b> | <b>Running (S=2)</b> | Upon product clearing the tracking path and the pneumatic actuator returning to its base position safely. | System automatically returns to monitoring the conveyor line for the next incoming items.        |
| <b>Normal Stop</b>     | <b>Running (S=2)</b> | <b>Idle (S=1)</b>    | When the operator presses the Stop Push Button (i_Stop_PB drops to FALSE).                                | Conveyor belt motor completely de-energizes, and operational registers                           |

|                       |                            |                    |                                                                                         |                                                                                                                                        |
|-----------------------|----------------------------|--------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|                       |                            |                    |                                                                                         | clear out to baseline.                                                                                                                 |
| <b>Emergency Trip</b> | <b>Any State (S=1,2,3)</b> | <b>Fault (S=4)</b> | NOT(i_EStop_Mon)<br>(Triggered immediately when the physical E-Stop button is slammed). | <b>Immediate hardware-level power cutoff to the motor contactor</b> bypassing software loops, and activating the amber warning beacon. |